NAME: CYBER SECURITY



Assignment Number 1

Kali Practice

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Roll No: 25L-2064

Section: BSCY-3A

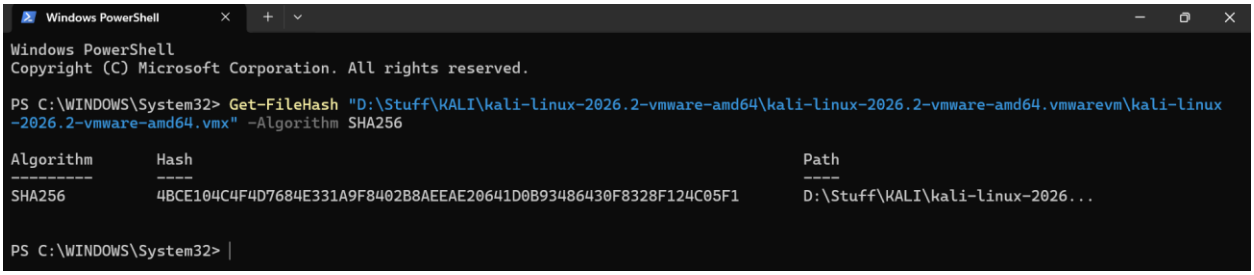
Friday 2nd October 2026

FAST School of, Computing, Lahore Campus

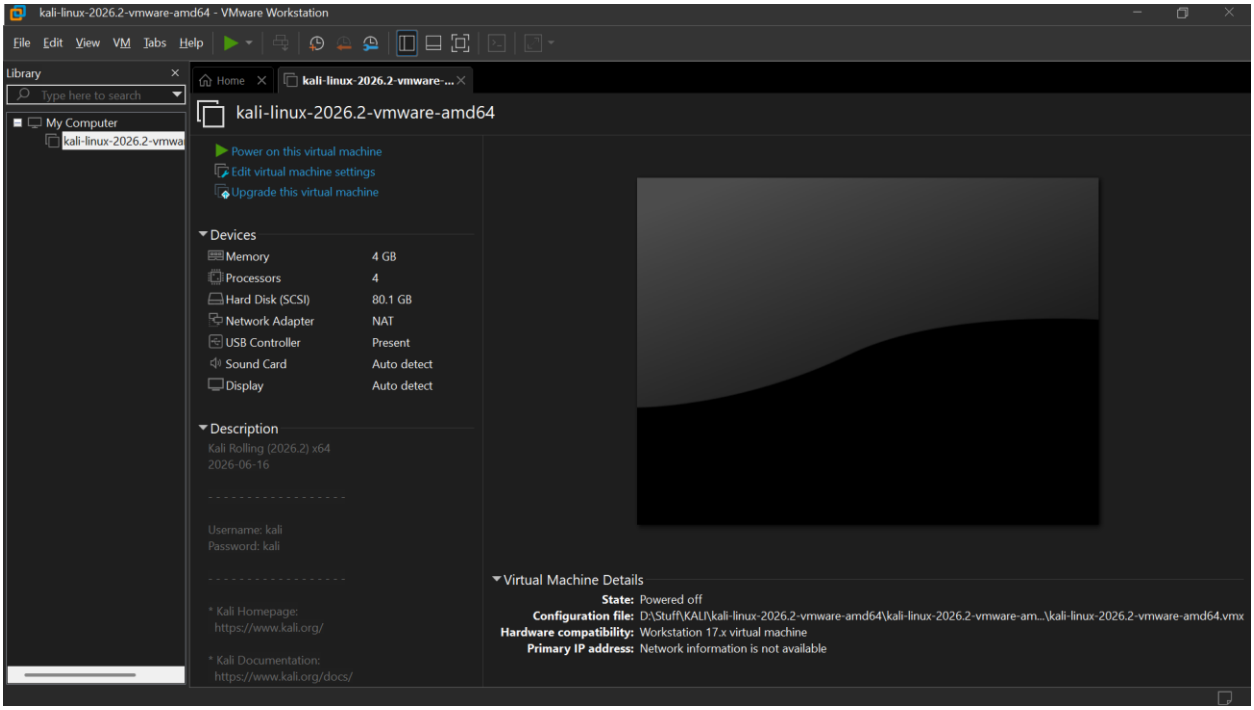
National University of Computer and Emerging Sciences

Lab Activity 2: Hash Verification

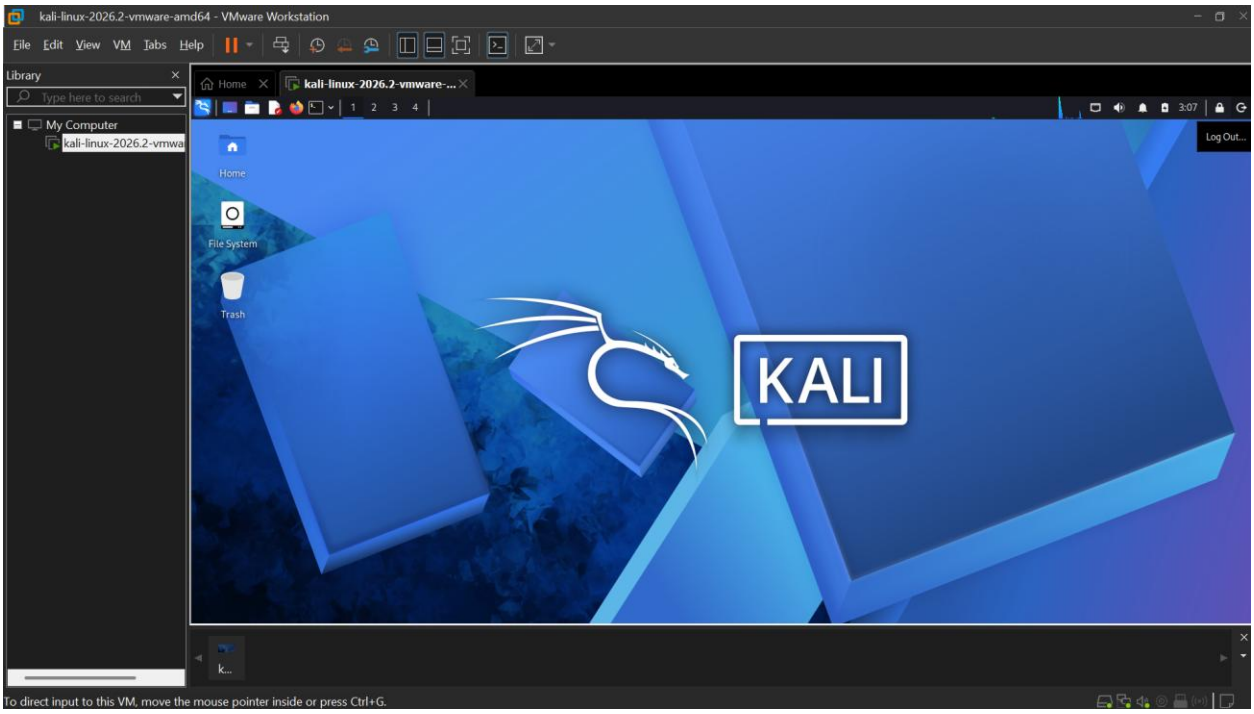
KALI IS INSTALLED.



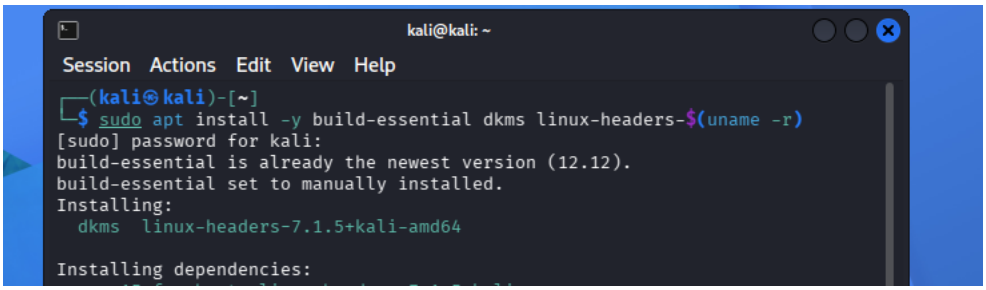
Lab Activity 3: VM Configuration

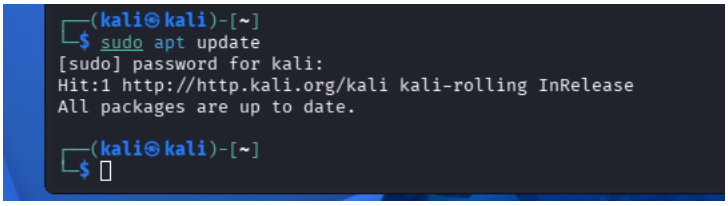


Kali Linux is Working

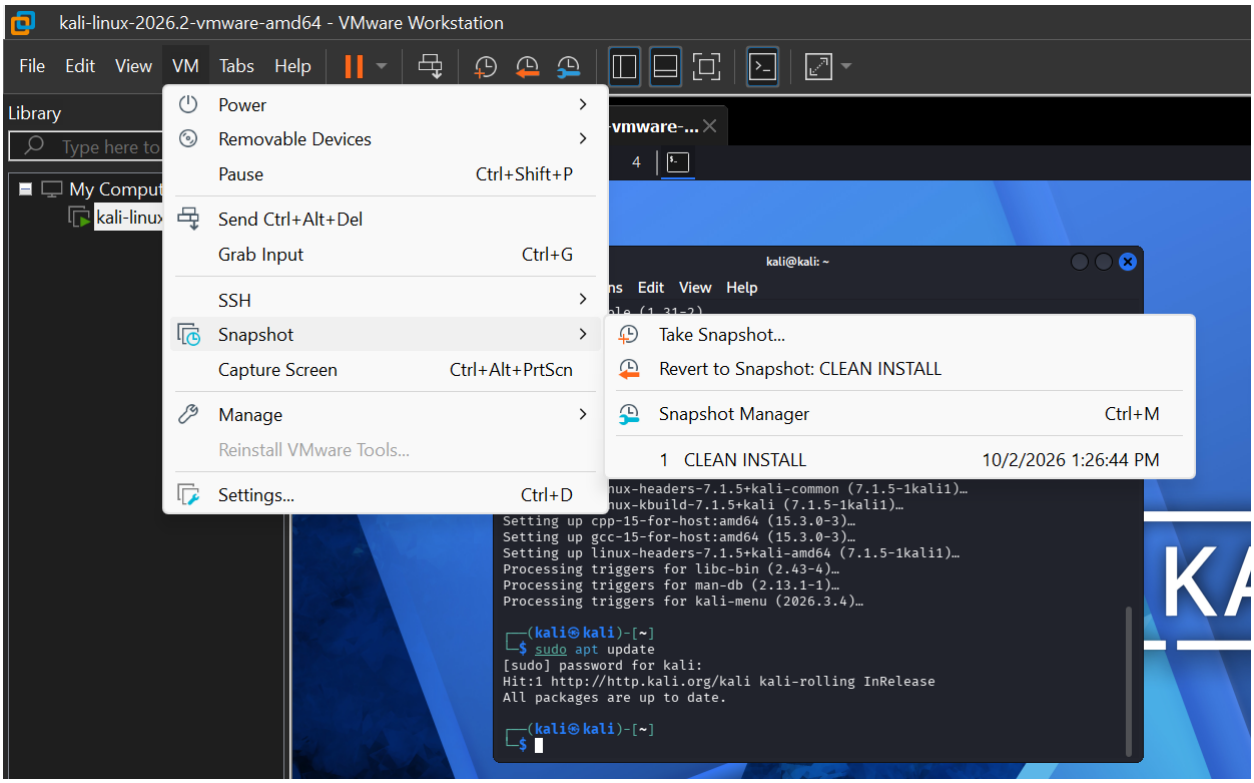


Using Commands.

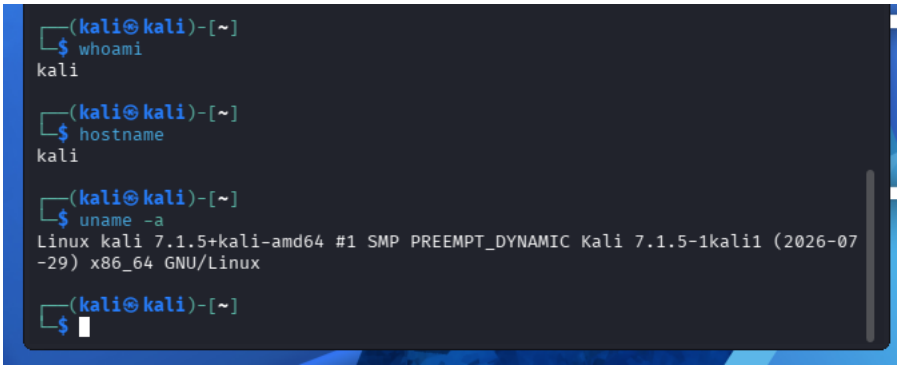




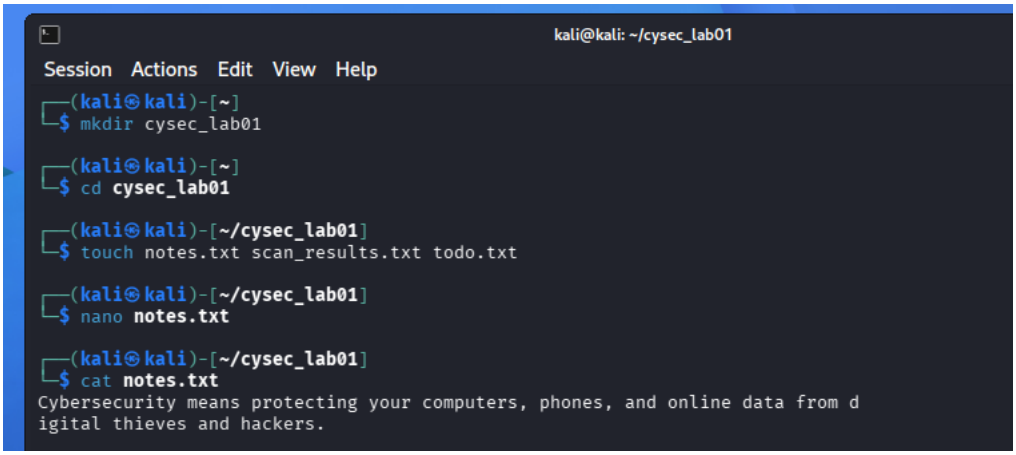
Made a SnapShot.



Checking whoami.



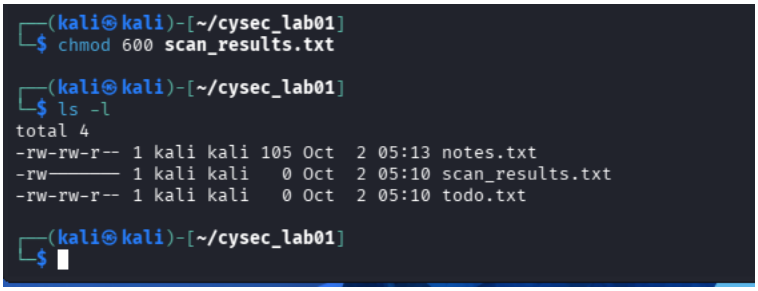
Performing Task. Making directory , and then changing directory , then making 3 text files and writing in it and reading it.



Finding .txt files.

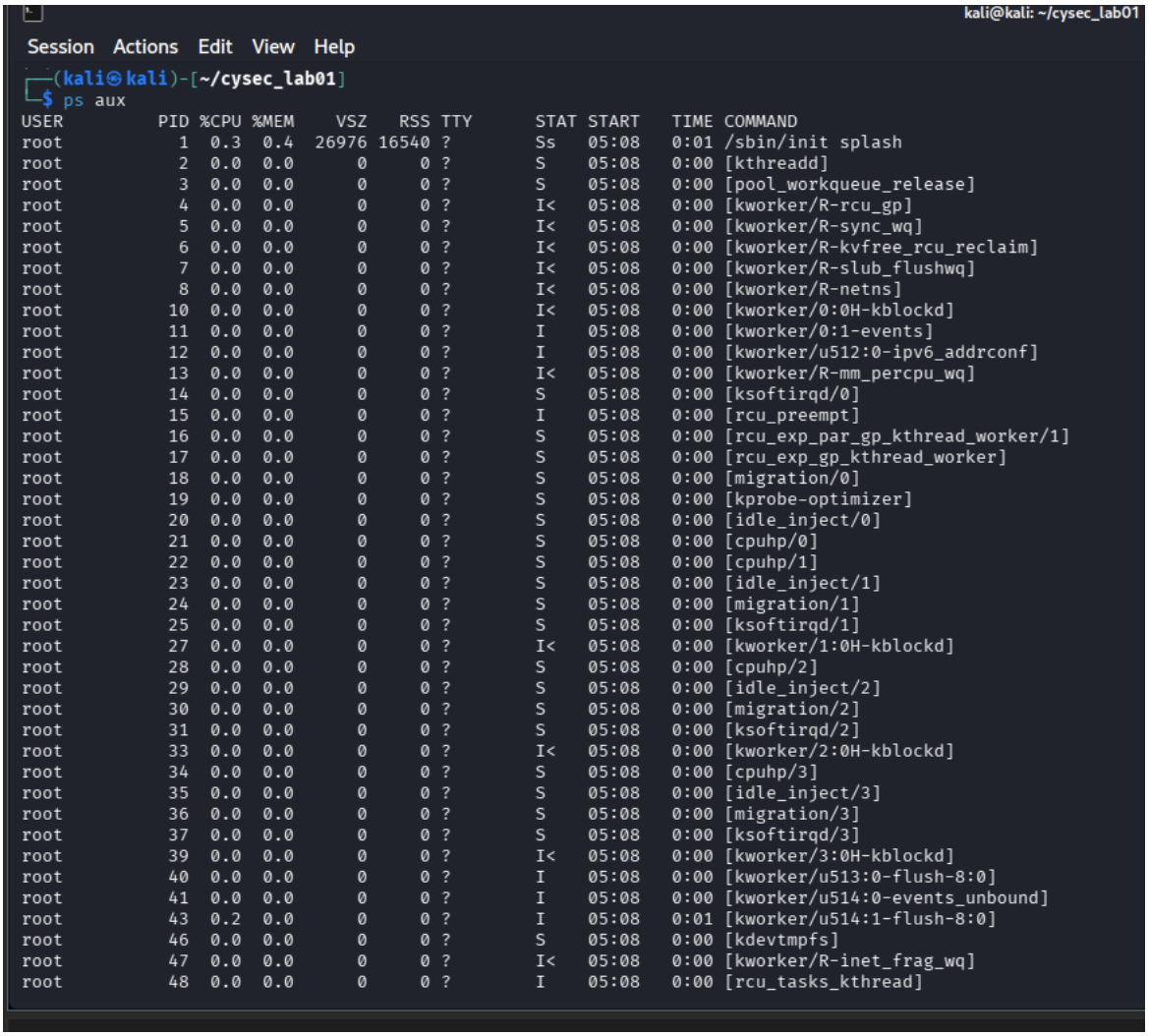


Restricting Permission.

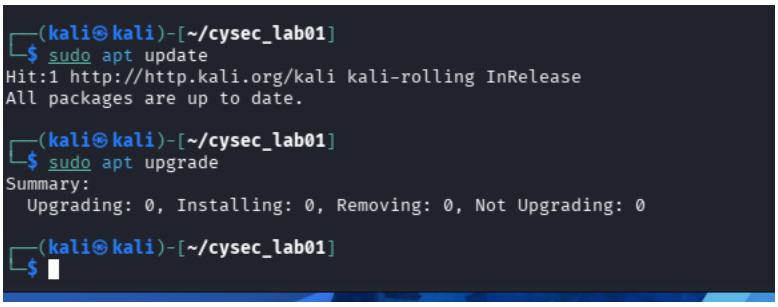


Processes:

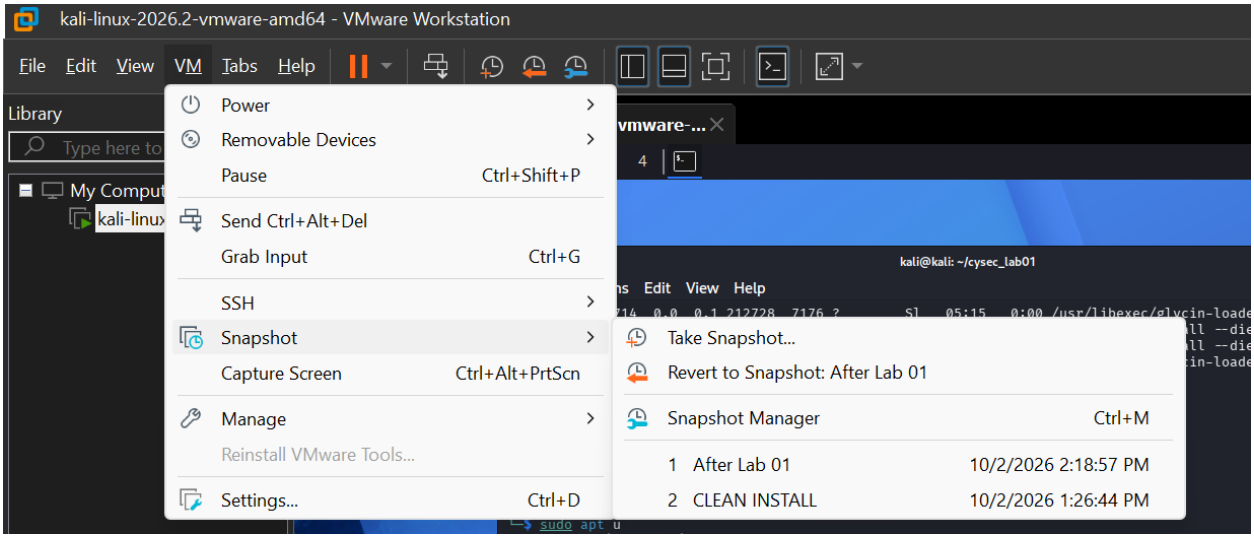
"/sbin/init splash is PID 1, kthreadd is PID 2."



Apt update and upgraded.



New SnapShot Created.



Multiple Choice Questions:

1. **a) Type-1 (bare-metal)**
2. **c) pwd**
3. **a) Read only**
4. **c) apt update**
5. **b) Save a restorable point-in-time state of the VM**

Short Questions:

1. We use VMs because they give us an isolated and safe environment. If we accidentally run malware or break the system during an experiment, it won't affect our main host machine, and we can just delete or restore the VM without any permanent damage.
2. The rm command is used to delete regular files, while rm -r deletes entire directories and everything inside them. We must be super careful with rm -rf because it forces the deletion of everything without asking for confirmation, which can destroy important system files if used in the wrong place.

3. The command chmod 640 report.txt sets specific permissions for a file:

- **6**: The owner can read (4) and write (2) files.
- **4**: The group can only read it.
- **0**: Others have zero access.

4. apt remove uninstalls a package but leaves its configuration files behind on your system. apt purge uninstalls the package and completely deletes its configuration files too, giving you a cleaner system.
5. Verifying the SHA256 checksum confirms that the ISO file you downloaded is original and exactly matches the official release. It ensures the file wasn't corrupted during download and hasn't been tampered with by an attacker.

Practical Self-Check Tasks

1. Directory/File and Permissions:
    - Create a directory: **mkdir my_folder**
    - Go into it and create a file: **cd my_folder** then **touch my_file.sh**
    - Change permissions so only I can execute it: **chmod 100 my_file.sh** (or chmod 700 my_file.sh if you also want to read and write access).
2. System Update Commands:
    - First, refresh the list: **sudo apt update**.
    - Then, install the updates: **sudo apt full-upgrade -y**.
3. Root vs Regular User and sudo:
    - The root user is the ultimate administrator with full power to change or delete anything on the system.
    - A regular user has limited permission to keep the system safe from accidental changes.
    - You use sudo when you are logged in as a regular user but need temporary admin privileges to run a specific command, like installing new software.